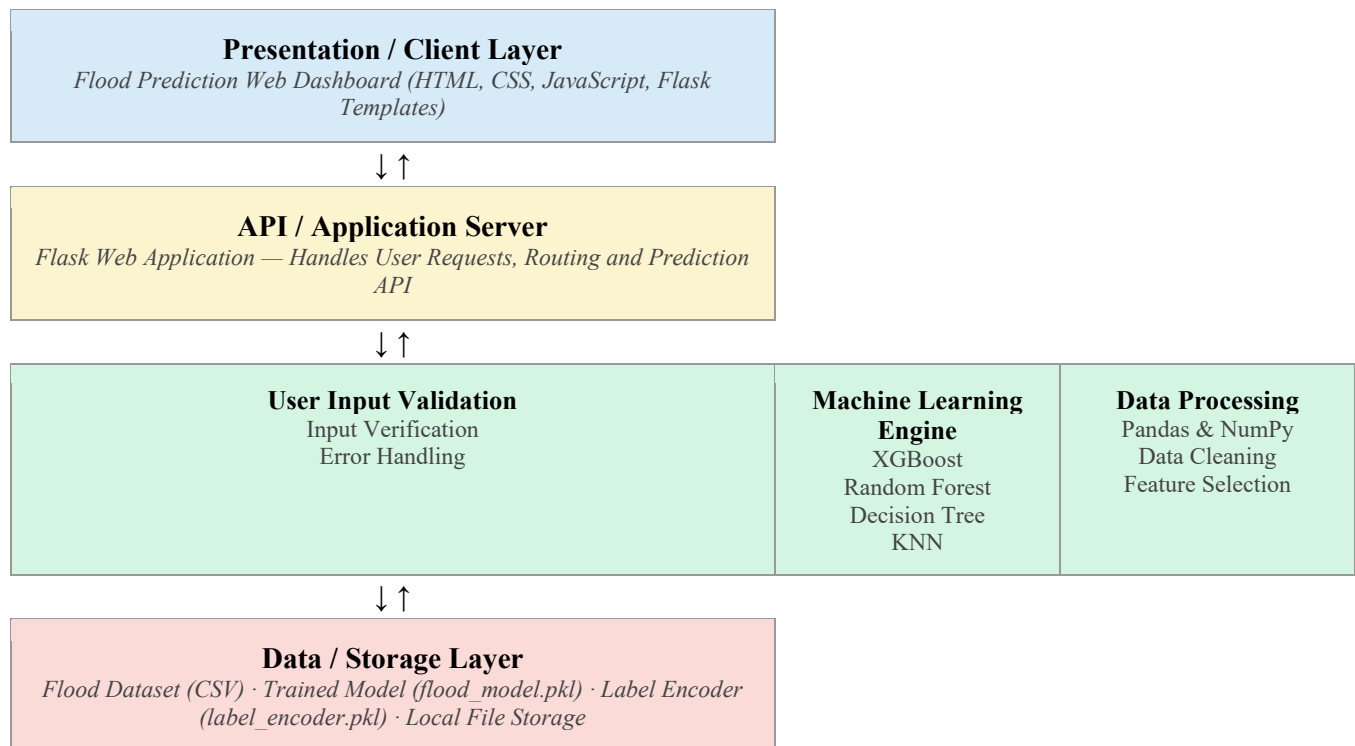


Solution Architecture

Date	15-07-2026
Team ID	
Project Name	Rising Waters
Maximum Marks	5 Marks

Solution Architecture Diagram



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides an interactive dashboard where users enter flood-related information and view prediction results with charts.	HTML5, CSS3, JavaScript, Chart.js
Application Layer (Flask Server)	Receives user input, processes requests, communicates with the machine learning model, and returns prediction results.	Python, Flask

Machine Learning Service	Loads the trained model, processes input data, predicts flood occurrence, and calculates flood risk.	Scikit-learn, XGBoost, Random Forest, Decision Tree, KNN
Data Processing Module	Cleans data, prepares features, splits datasets, and performs preprocessing before training.	Pandas, NumPy
Data / Storage Layer	Stores the flood dataset, trained machine learning model, and supporting files required for prediction.	CSV Dataset, Joblib (.pkl), Local File System

Technologies Used

- **Frontend:** HTML5, CSS3, JavaScript, Chart.js
- **Backend:** Python, Flask
- **Machine Learning:** Scikit-learn, XGBoost
- **Data Processing:** Pandas, NumPy
- **Model Storage:** Joblib (flood_model.pkl)
- **Dataset:** CSV
- **Development Tool:** Visual Studio Code
- **Version Control:** Git & GitHub